11

Lecture 11

Home Subscriber Server (HSS)

کارساز مشترکین خانگی (HSS)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate

PREREQUISITES
Lectures 9 and 10; basic probability; symmetric cryptography

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-11-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** the HSS as the union of HLR, AuC and IMS user-database functions, and **list** its interfaces.
- **LO2** **Explain** the subscription data model of the EPS and how the MME uses it.
- **LO3** **Trace** authentication-vector generation in the AuC and **explain** sequence-number management and re-synchronisation.
- **LO4** **Size** HSS storage and vector-request load, and **evaluate** the trade-off of requesting several vectors at once.
- **LO5** **Compute** the availability of geo-redundant HSS deployments.
- **LO6** **Relate** the HSS to the 5G UDM/AUSF/UDR and to data-layered (UDC) architectures.

2 Prerequisite knowledge

This lecture builds on the EPS-AKA procedure and key hierarchy of Lecture 9 and on the S6a application of Lecture 10. You should know what a block cipher and a message authentication code are, and be able to compute probabilities of independent events (for the availability models).

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
HSS	کارساز مشترکین خانگی	Master subscriber database of a 3GPP network (HLR + AuC + IMS data).
HLR	ثبت‌کنندهٔ موقعیت خانگی	2G/3G subscriber database reached over MAP.
AuC	مرکز احراز هویت	Stores the secret key K and generates authentication vectors.
IMSI	شناسهٔ بین‌المللی مشترک موبایل	Permanent subscriber identity (MCC + MNC + MSIN).
Authentication vector	بردار احراز هویت	(RAND, XRES, AUTN, K_{ASME}) for one EPS-AKA run.
SQN	شمارهٔ ترتیب	48-bit counter that proves the freshness of a vector.
AUTS	نشانهٔ همگام‌سازی مجدد	Token the USIM returns to re-synchronise SQN.
OPc	کلید پیکربندی اپراتور	Operator-variant value derived from OP and K for MILENAGE.
UDC / UDR	همگرایی دادهٔ کاربر / مخزن	Architecture separating application logic from a shared data repository.
Geo-redundancy	افزونگی جغرافیایی	Replicating a function across distant sites.

3 Introduction and motivation

A mobile network is, among other things, a very large and very available database. Every subscriber has a permanent identity, a secret key shared only with the SIM, a profile of allowed services and a current location; every attach, many handovers and every incoming call read or write some of these. In 2G and 3G these data lived in the Home Location Register (HLR) and the Authentication Centre (AuC); with IMS (Release 5) 3GPP introduced the *Home Subscriber Server* (HSS), which contains the HLR and AuC functions and adds the IMS user data [6]. In the EPS the HSS is the single subscriber authority for the MME, reached over Diameter S6a (Lecture 10).

The HSS is rarely discussed in textbooks because it does not move packets, yet an HSS outage stops every new attach in a country. Its design questions – how to keep 10^7–10^8 profiles consistent across sites, how to generate fresh authentication vectors at thousands per second, how to recover sequence numbers after a failure – are typical of carrier-grade distributed systems.

4 Fundamental concepts

4.1 Functions of the HSS

- **Identification and addressing**: IMSI, MSISDN, IMS public and private identities [7].
- **Authentication (AuC)**: stores K (and OPc) and generates authentication vectors [1], [2].
- **Subscription management**: stores the profile – allowed APNs, QoS, roaming and access restrictions – and pushes changes to the serving node.
- **Mobility management**: records the serving MME/SGSN/MSC so that the network can reach the subscriber, and cancels the old registration on a change of serving node.
- **IMS support**: stores the serving S-CSCF, IMS service profiles and initial filter criteria, and answers application servers [4].

4.2 Interfaces

Table 1. Main interfaces of the HSS.

Interface	Peer	Protocol	Use
S6a / S6d	MME / SGSN (EPS)	Diameter	Vectors, location update, subscription data [3]
Gr	SGSN (2G/3G)	MAP	Same functions for GPRS/UMTS [5]
C / D	GMSC / MSC-VLR	MAP	Call routing information, CS location update [5]
Cx	I-CSCF / S-CSCF	Diameter	IMS registration, S-CSCF assignment
Sh	Application server	Diameter	Transparent service data, notifications
SWx	3GPP AAA server	Diameter	Authentication for non-3GPP access (e.g. Wi-Fi)

4.3 Subscription data

Definition 1 • EPS subscription profile

The profile that the HSS sends in ULA (and later in IDR) contains, among others, the subscriber status, MSISDN, network-access mode, operator-determined barring, the subscribed UE-AMBR, access-restriction data (e.g. "no E-UTRAN"), the periodic TAU timer and an *APN-configuration profile*: a list of APN configurations, each with a context identifier, the APN name (Service-Selection), PDN type, the default bearer's QCI and ARP and the APN-AMBR [3].

The MME caches the profile while the subscriber is registered with it. The HSS remains the master: it changes the cached copy with Insert-Subscriber-Data (IDR) or Delete-Subscriber-Data (DSR), and withdraws it with Cancel-Location (CLR).

5 Detailed technical explanation

5.1 Authentication-vector generation

For each vector the AuC chooses a random RAND, takes the next sequence number SQN and evaluates five functions of K, RAND, SQN and AMF (Figure 1): f_1 gives the network authentication code MAC, f_2 the expected response XRES, f_3 and f_4 the cipher and integrity keys CK and IK,

and f_5 the anonymity key AK that conceals SQN [1]. The recommended algorithm set, MILENAGE, builds all five from one 128-bit block cipher (AES) and the operator constant OPc [8]. For E-UTRAN the HSS sets the *AMF separation bit* to 1 and derives K_{ASME} from CK and IK, binding the vector to the serving network [2]. The vector is

$$\text{AV} = \big(\text{RAND},\ \text{XRES},\ \text{AUTN},\ K_{\text{ASME}}\big), \qquad \text{AUTN} = (\text{SQN} \oplus \text{AK}) \,\|\, \text{AMF} \,\|\, \text{MAC}. \tag{1}$$

The secret K never leaves the AuC; the MME receives only vectors.

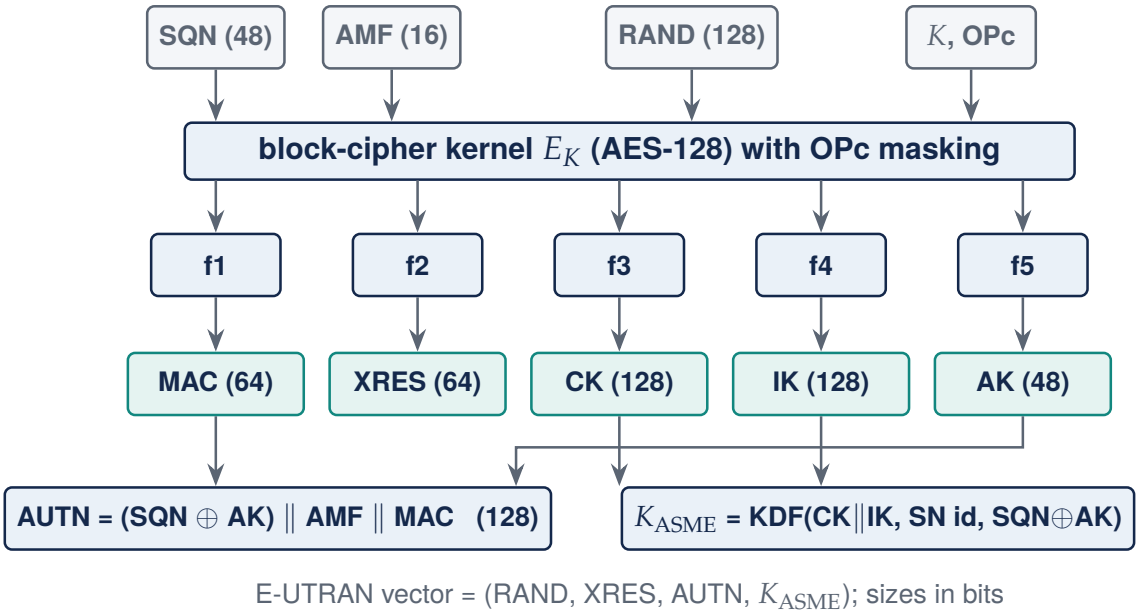


Figure 1. Authentication-vector generation in the AuC with the functions f_1–f_5.

5.2 Sequence numbers and freshness

The USIM accepts AUTN only if MAC is correct *and* SQN is fresh. TS 33.102 describes, in an informative annex, a scheme in which SQN is split into a 43-bit counter SEQ and a 5-bit index IND ($\text{SQN} = \text{SEQ} \,\|\, \text{IND}$) [1]. The USIM keeps an array of the highest SEQ accepted for each of the $2^5 = 32$ values of IND, and accepts a new SQN if its SEQ exceeds the stored value in the slot IND. The array lets vectors that were generated in one order be used in another – for example when two MMEs hold batches of vectors for the same subscriber – without spurious failures. An optional age check also rejects SQN values that are too far behind the highest accepted one.

If the SQN is not fresh, the USIM answers with a *synchronisation failure* carrying AUTS, which contains its own SQN concealed with a key derived like AK and protected by a MAC (MAC-S). The MME sends RAND and AUTS to the HSS in a new AIR (Re-Synchronization-Info); the AuC verifies MAC-S, sets its counter to the USIM's value and returns fresh vectors (Figure 4).

5.3 Mobility management in the HSS

On each ULR the HSS records the serving MME (Origin-Host and realm). If another MME was registered, the HSS sends it CLR so that it deletes its context; this guarantees one serving node per subscriber. An MME that deletes a long-inactive subscriber sends Purge-UE (PUR), after which the HSS marks the subscriber as purged. After a restart, an HSS that may have lost registrations can send Reset (RSR) to the MMEs, which then refresh the affected subscribers at their next contact

[3].

5.4 Implementation: front-ends, data store and UDC

Commercial HSSs separate stateless *front-ends* that terminate MAP and Diameter and run the application logic from a replicated, mostly in-memory *data store*. 3GPP generalised this as *User Data Convergence* (UDC): HLR/HSS, AuC and other front-ends share one User Data Repository (UDR) over the Ud interface, so that a subscriber has one record instead of a copy per node (Figure 2). The 5G core keeps the same split: UDM and AUSF are front-ends, UDR is the repository (Lectures 18–19). Because the HSS is a single point of control, the data store is replicated across geographically separated sites.

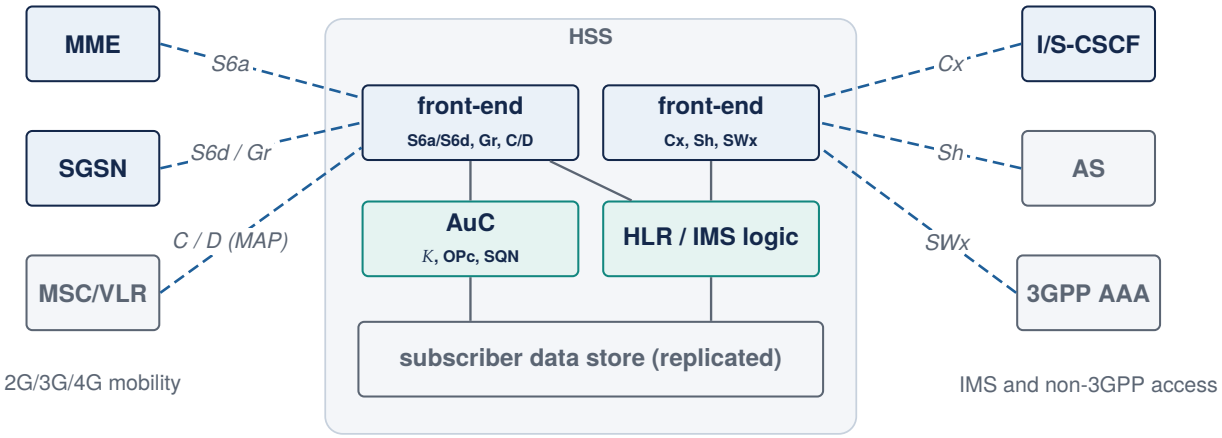


Figure 2. Functional view of the HSS with its interfaces and the front-end/data-store split.

6 Mathematical models

6.1 Sequence-number lifetime

Model. With a b-bit SEQ and one authentication every τ seconds, the counter lasts

$$T_{\text{life}} = 2^b \, \tau. \tag{2}$$

Example. $b = 43$, one authentication per minute: $T_{\text{life}} = 8.8 \times 10^{12}$ min $\approx 1.7 \times 10^7$ years. Exhaustion is not a concern; the counter is large so that schemes that skip values (for example a SEQ derived from a clock) remain possible.

6.2 Storage

Model. N subscribers with a mean profile size s, replicated R times, with a fill factor η (fraction of memory that may be used):

$$M = \frac{N \, s \, R}{\eta}. \tag{3}$$

Example. $N = 10^7$, $s = 4$ kB, $R = 3$, $\eta = 1$: $M = 120$ GB – small enough to keep in memory on a few servers. Storage is therefore not the hard problem; consistency and availability are.

6.3 Vector requests

Model. If authentications occur at rate λ_{auth} and the MME requests n vectors per AIR and uses all of them, the AIR rate is

$$\lambda_{\text{AIR}} = \frac{\lambda_{\text{auth}}}{n}. \tag{4}$$

Example. 10^7 subscribers authenticated 10 times a day: $\lambda_{\text{auth}} = 1157\,\text{s}^{-1}$; $n = 5$ gives $\lambda_{\text{AIR}} = 231\,\text{s}^{-1}$.

Limitation. Vectors left unused when the subscriber moves to another MME are wasted, and stored vectors consume MME memory, so n is usually small (Exercise 4).

6.4 Availability of redundant sites

Model. Each site is available with probability a (unavailability $u = 1 - a$), independently of the others. If service needs at least k of N sites,

$$A_{k\text{-of-}N} = \sum_{i=k}^{N} \binom{N}{i} a^i (1-a)^{N-i}. \tag{5}$$

Special cases: $A_{1\text{-of-}2} = 1 - u^2$; $A_{2\text{-of-}3} = 1 - (3u^2 - 2u^3)$; $A_{2\text{-of-}2} = a^2$.

Example. $a = 0.999$ (8.76 h downtime per year): two sites each able to carry the full load give $u^2 = 10^{-6}$, i.e. 31.5 s per year; two sites that are both needed give 17.5 h per year (Figure 3).

Assumptions and interpretation. Independence is the critical assumption: software bugs, configuration errors and replication faults hit all sites at once, and in practice dominate HSS outages. The model gives an upper bound on what redundancy can achieve, not a prediction.

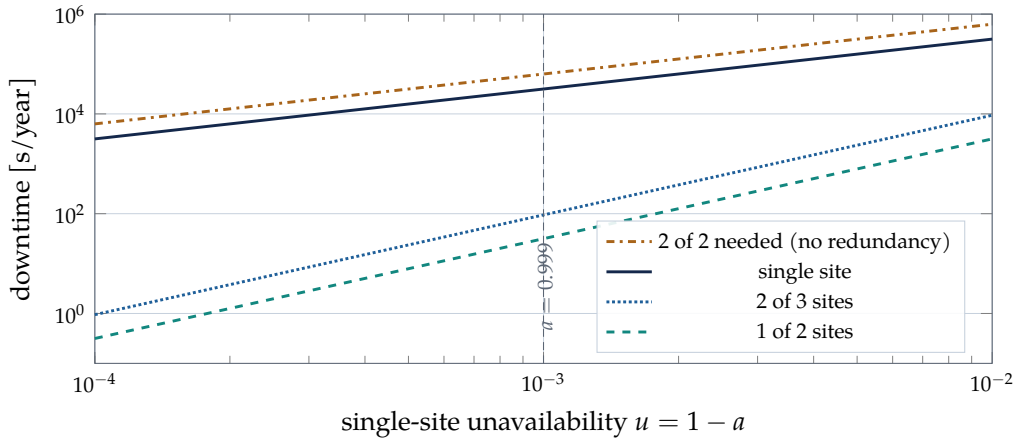


Figure 3. Annual downtime as a function of single-site unavailability for four deployment options (independent failures assumed).

7 Network architecture and diagrams

Figure 2 shows the HSS between the circuit and packet core of 2G/3G, the EPS and the IMS. The same subscriber record serves all of them: a VoLTE subscriber is authenticated for E-UTRAN over

S6a, registered in the IMS over Cx, and reached by an MSC over MAP if the call falls back to the circuit domain. In the EPS, the HSS has exactly one control-plane peer type for mobility, the MME, and is usually reached via DRAs (Lecture 10).

8 Step-by-step procedures

Procedure 1 • Generating an E-UTRAN vector in the AuC

1. Receive AIR with the IMSI and the serving-network identity (Visited-PLMN-Id).
2. Look up K and OPc; generate a fresh 128-bit RAND; take the next SQN (SEQ and IND).
3. Compute MAC $= f_1$, XRES $= f_2$, CK $= f_3$, IK $= f_4$, AK $= f_5$ with AMF separation bit $= 1$.
4. Form AUTN with Eq. (1); derive K_{ASME} from CK, IK, the serving-network identity and SQN$\oplus$AK.
5. Return the vectors in AIA; the HSS discards CK and IK.

Procedure 2 • Sequence-number re-synchronisation

1. The USIM verifies MAC, recovers SQN and finds it not fresh; it computes AUTS and the UE sends Authentication Failure (synch failure).
2. The MME discards its remaining vectors for this subscriber and sends AIR with Re-Synchronization-Info = RAND $\|$ AUTS.
3. The AuC verifies MAC-S, sets its SQN to the USIM's value and generates new vectors.
4. The MME restarts authentication with a new vector; it now succeeds.

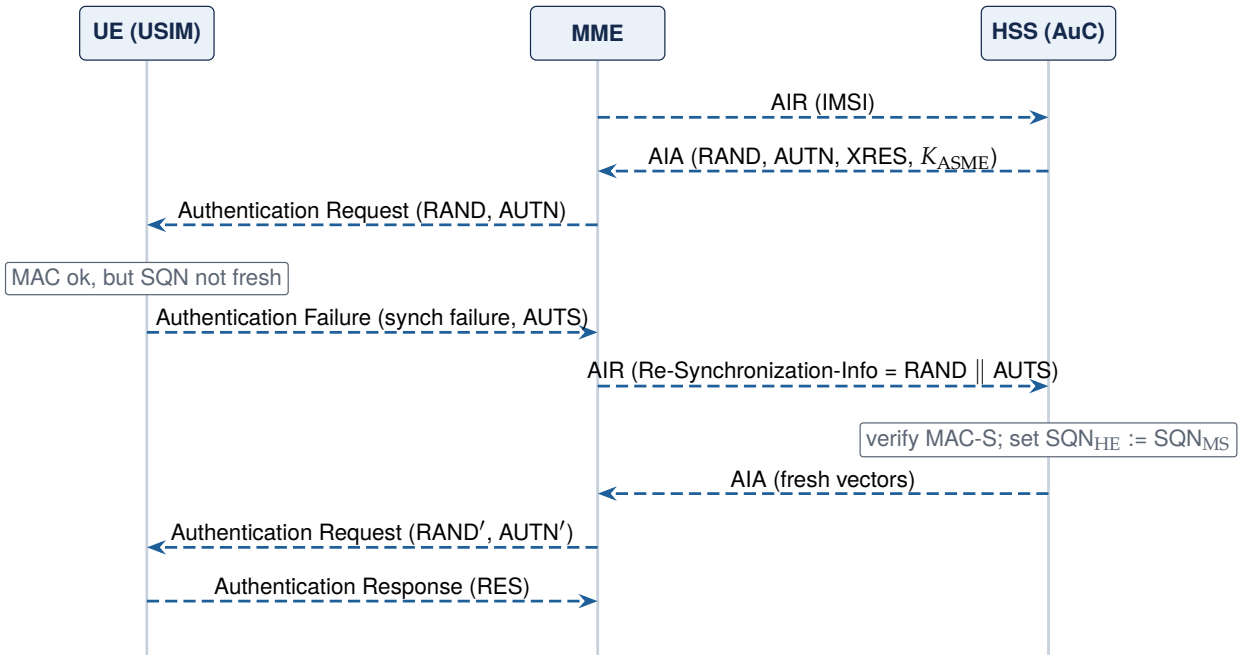


Figure 4. Sequence-number re-synchronisation (Procedure 2).

9 Worked examples

Worked example 1 • Size of an AIA

Problem. Compute the size of the Authentication-Info AVP carrying five E-UTRAN vectors (all AVPs vendor-specific, optional Item-Number omitted).
Solution. Using the AVP length rule of Lecture 10 with a 12-byte header: RAND $12+16=28$, XRES (64 bits) $12+8=20$, AUTN $12+16=28$, K_{ASME} $12+32=44$. One E-UTRAN-Vector: $12+120=132$ bytes; five vectors in the grouped Authentication-Info: $12+5\times 132=672$ bytes.

Worked example 2 • Restore from backup

Problem. An HSS of 10^7 subscribers is restored from a one-day-old backup, so its SQN counters are behind the USIMs'. If all subscribers authenticate within the next hour, how much extra S6a load is created?
Solution. Each subscriber's first authentication fails with a synchronisation failure and needs one extra AIR: $10^7/3600=2778$ extra AIR per second for an hour – on top of normal traffic and comparable to it. Operators avoid this by replicating SQN updates continuously rather than relying on backups, or by advancing restored counters by a safe margin.

Worked example 3 • Choosing a redundancy scheme

Problem. Sites have $a=0.999$. Compare one site, two sites each able to carry the full load, and three sites of which any two can carry the load.
Solution. From Eq. (5): one site $u=10^{-3}$ (8.76 h/year); 1-of-2 $u^2=10^{-6}$ (31.5 s/year); 2-of-3 $3u^2-2u^3\approx 3.0\times 10^{-6}$ (94.5 s/year). The 2-of-3 design is three times less available than 1-of-2 but needs only 50 % capacity per site instead of 100 %, i.e. 150 % total instead of 200 %.

10 Technology comparison

Table 2. Subscriber-data functions across generations.

	HLR/AuC (2G/3G)	HSS (4G, IMS)	UDM/AUSF/UDR (5G)
Protocol	MAP over SS7/SIGTRAN	Diameter (S6a, Cx, Sh)	HTTP/2 SBI (Nudm, Nausf, Nudr)
Authentication	GSM triplets, UMTS quintets	EPS vectors with K_{ASME}	5G-AKA, EAP-AKA′
Identity privacy	TMSI only	TMSI/GUTI only	Concealed SUPI (SUCI)
Data model	Per-node records	Profile per access + IMS	Common UDR, NF front-ends
Discovery	Global titles	Realms, DRAs	NRF

HSS versus visitor databases. The HSS is the master copy; the MME (like the VLR in 2G/3G) caches what it needs for its subscribers. Caching keeps the HSS out of most procedures – a TAU within the same MME needs no S6a message – at the price of keeping copies consistent with IDR/DSR/CLR.

11 Key takeaways

Key takeaways

- HSS = HLR + AuC + IMS user data; the single subscriber authority, reached over S6a, Cx, Sh, SWx and MAP.
- The AuC keeps K secret and issues vectors (RAND, XRES, AUTN, K_{ASME}) built with f_1–f_5 (MILENAGE).
- SQN = SEQ $\|$ IND proves freshness; failures are repaired with AUTS re-synchronisation.
- The HSS records the serving MME, cancels old registrations and pushes profile changes.
- Storage is small ($\approx$100 GB for 10^7 users); availability and consistency are the hard parts, modelled by k-of-N redundancy with independence as the key assumption.
- The front-end/repository split (UDC) leads directly to UDM/AUSF/UDR in 5G.

12 Exercises

Exercise 1 • Level 1 – conceptual • Master and cache

Which subscriber data are stored permanently in the HSS and which are cached in the MME? Which messages keep the cache consistent?

Exercise 2 • Level 1 – conceptual • Realm from IMSI

For IMSI 432101234567890 with a two-digit MNC, write the EPC home-network realm that a visited MME uses as Destination-Realm.

Exercise 3 • Level 2 – numerical • Storage dimensioning

Compute the memory needed for 5×10^7 subscribers with a 6 kB profile, three replicas and a fill factor of 0.7.

Exercise 4 • Level 2 – numerical • How many vectors per request?

With $\lambda_{\mathrm{auth}} = 1157\,\mathrm{s}^{-1}$, compute the AIR rate for $n = 1, 3, 5$. If a subscriber uses on average two vectors per MME visit before moving to another MME, what fraction of generated vectors is wasted for $n = 5$?

Exercise 5 • Level 3 – analytical • Redundancy design

Derive the expressions for $A_{2\text{-of-}3}$ and $A_{1\text{-of-}2}$ from Eq. (5). Then suppose that, besides independent failures, a common-mode failure (e.g. a faulty software release) disables all sites with probability $p_c = 10^{-5}$. Recompute the 1-of-2 downtime and comment.

Exercise 6 • Level 3 – analytical • The IND array

MME A holds an unused vector with IND = 7. The HSS keeps issuing vectors to MME B, incrementing IND cyclically. After how many newly issued vectors can the vector held by MME A become unacceptable, and under which condition does it actually fail?

Solutions

Solution to Exercise 1

Permanent in the HSS: IMSI, MSISDN, K/OPc and SQN (AuC), the subscription profile and the identity of the serving MME. Cached in the MME: the profile (from ULA), unused vectors and the security context. Consistency: IDR/DSR (profile changes), CLR (withdrawal), PUR (MME deletes the subscriber), RSR (HSS restart).

Solution to Exercise 2

MCC = 432, MNC = 10 (padded to three digits as 010), so the realm is

`epc.mnc010.mcc432.3gppnetwork.org`

[7]. The MNC length is not encoded in the IMSI; the MME needs it from configuration.

Solution to Exercise 3

$M = 5 \times 10^7 \times 6 \times 10^3 \times 3/0.7 = 1.29 \times 10^{12}$ bytes ≈ 1.3 TB – distributed over a cluster, still feasible in memory.

Solution to Exercise 4

$\lambda_{\text{AIR}} = 1157,\,386,\,231\ \text{s}^{-1}$. With $n = 5$ each visit triggers one AIR and uses two of five vectors: three of five (60 %) are wasted, so the AuC computes $5/2 = 2.5$ times as many vectors as needed. With $n = 3$, one AIR per visit and 33 % waste. The best n matches the typical number of authentications per visit.

Solution to Exercise 5

$A_{1\text{-of-}2} = 2a(1-a) + a^2 = 1 - (1-a)^2$. $A_{2\text{-of-}3} = 3a^2(1-a) + a^3$, so $1 - A = 1 - 3a^2 + 2a^3 = 3u^2 - 2u^3$ with $u = 1 - a$. With a common mode: $1 - A \approx p_c + u^2 = 10^{-5} + 10^{-6} = 1.1 \times 10^{-5}$, i.e. 347 s per year instead of 31.5 s. The common-mode term dominates by a factor of ten: beyond two sites, staged software roll-outs and configuration discipline improve availability more than extra hardware.

Solution to Exercise

IND takes 32 values, so the next vector with IND $= 7$ (and a higher SEQ) is issued after 32 new vectors. The vector held by MME A fails only if that newer vector is *used* by the USIM before MME A uses its own: the USIM then stores the higher SEQ in slot 7 and rejects the older one. Up to 31 newer vectors can therefore be used in any order without affecting it.

13 Further reading

The HSS is defined in TS 23.002 [6]; identities and realms in TS 23.003 [7]. The authentication framework and the SQN management annex are in TS 33.102 [1], the EPS-specific parts in TS 33.401 [2], and MILENAGE in TS 35.206 [8]. S6a is specified in TS 29.272 [3], MAP in TS 29.002 [5]. Olsson et al. [9] describe the HSS from a deployment perspective.

References

[1] 3GPP, *3G security; security architecture*, TS 33.102, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/33102.htm.

[2] 3GPP, *3GPP system architecture evolution (SAE); security architecture*, TS 33.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/33401.htm.

[3] 3GPP, *Evolved packet system (EPS); mobility management entity (MME) and serving GPRS support node (SGSN) related interfaces based on Diameter protocol*, TS 29.272, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29272.htm.

[4] 3GPP, *IP multimedia subsystem (IMS); stage 2*, TS 23.228, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23228.htm.

[5] 3GPP, *Mobile application part (MAP) specification*, TS 29.002, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/29002.htm.

[6] 3GPP, *Network architecture*, TS 23.002, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23002.htm.

[7] 3GPP, *Numbering, addressing and identification*, TS 23.003, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23003.htm.

[8] 3GPP, *Specification of the MILENAGE algorithm set: An example algorithm set for the 3GPP authentication and key generation functions; document 2: Algorithm specification*, TS 35.206, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/35206.htm.

[9] M. Olsson, S. Sultana, S. Rommer, L. Frid, and C. Mulligan, *EPC and 4G Packet Networks: Driving the Mobile Broadband Revolution*, 2nd ed. Academic Press, 2013.